

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

Computer Vision – 21CSE454T

UNIT 2: SHAPES AND REGIONS | SESSION 9: REGION DESCRIPTORS

SLO 2: Apply region descriptors to perform shape analysis and object classification in images.

FEATURE CLUE GAME SOLUTIONS

- **Case 1:** Area is large | Centroid is at the center | Width $\approx$ Height
Predicted Shape: **Large Circle / Square (Symmetrical Shape)**
- **Case 2:** Area is moderate | Height is much larger than width | Centroid lies along the vertical center
Predicted Shape: **Elongated Vertical Rectangle / Rod / Bar**
- **Case 3:** Area is small | Three major corners | Centroid near geometric center
Predicted Shape: **Triangle**

SHAPE INVESTIGATOR SOLUTIONS

Object	Area	Aspect Ratio	Eccentricity	Predicted Class
A	120	1 : 1	Low	Circle / Square (Symmetrical)
B	180	3 : 1	High	Elongated Rectangle / Bar
C	90	1 : 1	Medium	Rounded / Regular Polygon

Question: Which region descriptor helped most in distinguishing Object B?

Eccentricity (and Aspect Ratio) helped most in distinguishing Object B by quantifying its high degree of elongation.